COMMERCIAL ITEM DESCRIPTION

HOLDERS, ELECTRODE, SHIELDED METAL, ARC WELDING, TONG TYPE

The General Services Administration has authorized the use of this commercial item description in preference to type II, class A, sizes 1, 2, and 3 of A-A-51161.

1. ABSTRACT

1.1 Abstract. This description covers five sizes of hand held electrode holders of the tong type which are fully insulated and are used in the shielded metal arc welding (SMAW) process.

2. CLASSIFICATION

2.1 Classification. The holders shall be of one type, and the following sizes. The size to be furnished shall be as specified (see 7.1).

Size 200 - 200 amperes
Size 250 - 250 amperes
Size 300 - 300 amperes
Size 400 - 400 amperes
Size 500 - 500 amperes

3. SALIENT CHARACTERISTICS

3.1 Design. The holder shall be of the tong type, with a hinge and spring connecting the upper and lower jaws. The lower jaw shall have grooves to hold the electrode in a predetermined fixed position during welding operations. The holder shall be insulated in such a manner as to prevent current flow when placed on a flat conductive surface. There shall be no exposed metal parts in the handle or upper tong lever which the operator may contact. The holder shall be similar to that shown in figure 1.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5213.

AMSC N/A FSC 3439

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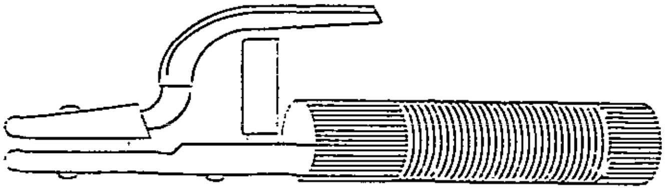


FIGURE 1. Electrode holder, tong type.

3.2 Holder body. The electrical current carrying body of the holder shall be made of either manganese bronze, brass, or copper. The electrical conducting components of the electrode holders shall have a minimum International Annealed Copper Standard (IACS) volumetric electrical conductivity of 20 percent at 68 degrees Fahrenheit when tested in accordance with ASTM B 193. The hardness of the holders metal body shall be not less than 50 and not more than 70 Rockwell on the "B" scale when tested in accordance with ASTM E 18. The body shall have provisions for attaching an insulated handle, of not less than 5 inches in length.

3.3 Insulation. The holder shall be insulated with a material which has the strength and toughness to withstand the intense heat produced during the welding process.

3.4 Dimensions and tolerances. The dimensions and tolerances of the holders supplied shall conform to table I.

TABLE I. Holder dimensions and tolerances.

Size	Ampere rating	Holder size (max) (inches)	Cable size (max) (AWG) 1/	Holder weight (ounces) 2/	Holder length (inches)3/
200	200	5/32	1/0	15	8.5
250	250	3/16	2/0	15	9.5
300	300	1/4	3/4	25	10.0
400	400	1/4	4/0	27	12.0
500	500	3/8	4/0	35	13.5

1/ The holder shall be capable of holding cable up to the size stated.
2/ The holder shall not exceed stated weight.
3/ The tolerance shall be ± 1/2 inch.

3.5 Impact requirement. The holder shall withstand a six foot drop on a hard surface without being damaged to the extent there would be a safety hazard to the user.

3.6 General requirements.

3.6.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.6.2 Environmental protection. During the manufacture of the electrode holders, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.6.3 Reference documents. The documents referenced in this document shall be the issues in effect on the data of issuance of the invitation for bid or request for proposal.

4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall certify and maintain objective quality evidence that the product offered meets this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provision of the contract.

6. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

6.1 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract order.

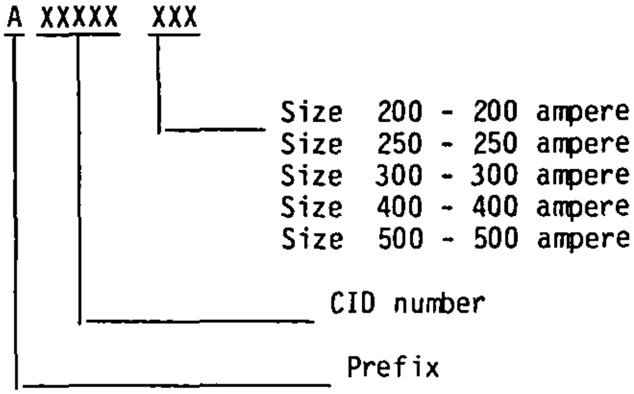
7. NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Size and quantity required.

7.2 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



7.3 Source for documents referenced.

FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.

ASTM E 18 and ASTM B 193 are available from American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103-1187.

7.4 Preparing activity. This document was prepared at the Defense Industrial Plant Equipment Center, 2163 Airways Blvd., Memphis, TN 38114-5213, phone number, commerical 901-775-6247, DSN 683-6247.

MILITARY INTEREST:

Custodians

Army - AL
Navy - SH
Air Force -99

Review Activities

Air Force - 84
DLA - GS

User Activities

Navy - MC, YD, AS

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3439 -0807

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-55013	2. DOCUMENT DATE *(YYMMDD)* 92 02 04

3. DOCUMENT TITLE
Holders, Electrode, Shielded Metal Arc Welding Tong Type

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*

b. ORGANIZATION

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(1) Commercial
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8. PREPARING ACTIVITY

a. NAME
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSG

b. TELEPHONE *(Include Area Code)*
(1) Commercial
(901)-775-6247
(2) AUTOVON
683-6247

c. ADDRESS *(Include Zip Code)*
2163 Airways Blvd.
Memphis, TN 38114-5213

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